AXIS^2: TWO-DIMENSIONAL LATENT GENERATION FOR LANGUAGE MODELING

Anonymous authors
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ABSTRACT

Language generation has explored several decoding paradigms. Autoregressive models generate tokens sequentially, while diffusion and masked-token models generate through iterative refinement. More recent latent and hierarchical approaches further move generation away from direct next-token prediction by operating on alternative representations or semantic scales. Despite these differences, generation is typically organized along a single decoding progression. We introduce AXIS^2, which treats generation progress as two-dimensional rather than collapsing it into a single decoding trajectory. Along the first axis, the model progresses through a sequence of representations, ($\mathbf{q}_1 \to \mathbf{q}_2 \to \cdots \to \mathbf{q}_K$), where later representations contain progressively more information needed to recover the final output. Along the second axis, generation within each representation level ($\mathbf{q}_k$) proceeds over multiple rounds: each round generates a subset of the remaining variables conditioned on those generated in previous rounds. Together, the two axes perform complementary refinement in latent space: one progressively increases the information represented about the output, while the other strengthens dependencies among latent variables at the same level. Their combination yields increasingly detailed and better-coordinated latent representations before they are decoded into the final output. Across multiple text generation benchmarks, AXIS^2 consistently improves generation quality over strong autoregressive, diffusion-based, and latent-space baselines while substantially reducing the number of full-model forward evaluations.

1 INTRODUCTION

Human language production is not simply a process of making one word-level decision after another. Classical psycholinguistic accounts characterize speech production as a hierarchical and incremental process: speakers can form higher-level intentions about what to communicate before the exact lexical realization is determined, while continuing to specify upcoming content as an utterance unfolds. (**???**). For example, when answering a complex question in conversation, a speaker may already have a rough idea of the main point to convey before deciding the exact words used to express it. As the utterance unfolds, upcoming content is further specified while the current content is being realized. Nor is this planning completed before speaking begins; we continue organizing upcoming content while realizing the current one. Thus, although language is ultimately expressed as a linear sequence of words, its production naturally involves decisions at multiple levels of abstraction: higher-level decisions determine *what to say*, while lower-level decisions progressively determine *how to say it*.

Recent work has begun to move beyond standard next-token prediction. On one hand, Multi-Token Prediction (MTP) predicts multiple future tokens at once, increasing parallelism while retaining token-level prediction targets (Gloeckle et al., 2024). On the other hand, diffusion language models relax strict autoregressive ordering by updating many tokens in parallel through iterative denoising or masked refinement (Li et al., 2022; Sahoo et al., 2024; Nie et al., 2025b). Yet these approaches still operate primarily over variables at the final token resolution: they change how many tokens can be predicted or refined together, but do not explicitly organize generation across multiple levels of representation. This leaves open a complementary question: can language generation itself be structured hierarchically, so that coarse content organization and fine-grained linguistic realization are jointly incorporated into the generative process?

In this work, we introduce **AXIS**^{2}, a two-dimensional coarse-to-fine framework for language generation . Along the first dimension, AXIS^2 generates progressively finer representations instead of directly predicting final tokens, allowing higher-level content structure to be established before increasingly specific semantic and linguistic details are resolved. Along the second dimension, AXIS^2 does not require all variables at a given resolution to be generated simultaneously; it progressively commits subsets of them and conditions subsequent predictions on what has already been resolved, preserving dependencies within each resolution. Together, these two dimensions separate high-level content organization from low-level linguistic realization while retaining local dependency modeling. They also substantially shorten the expensive sequential computation path: a single full-model forward pass can advance many generation variables, reducing repeated model execution and parameter movement. This is particularly valuable for low-batch inference, where repeated full-model evaluations directly contribute to latency and memory-bandwidth cost.

We evaluate AXIS^2 on multiple text generation benchmarks against autoregressive, diffusion-based, and other parallel-generation baselines. Across these settings, AXIS^2 achieves a favorable trade-off between generation quality and efficiency ...

TODO: show efficiency and quality results.

2 Related Work

Token-Level Language Generation. Most language models generate directly at the output-token level, while differing in how predictions are ordered and parallelized. Autoregressive models generate one token at a time, multi-token prediction predicts several future tokens jointly (Gloeckle et al., 2024), and masked or diffusion-based models iteratively refine multiple token states (Ghazvininejad et al., 2019; Austin et al., 2021; Li et al., 2022; Lou et al., 2024; Sahoo et al., 2024). Recent work scales masked diffusion, combines diffusion with block-wise autoregression, or reduces sampling cost (Nie et al., 2025a;b; Arriola et al., 2025; Sahoo et al., 2025). HDLM further introduces coarse-to-fine semantic states within discrete diffusion (Zhou et al., 2025). These works primarily study how token-level generation can be ordered, parallelized, or iteratively refined, whereas AXIS^2 changes the generative representation itself, generating a hierarchy of latent representations before decoding them into tokens.

Latent Representations in Language Generation. Latent representations may either augment token generation or themselves serve as generative variables. NextLat and HiLP retain token-level generation while introducing latent-prediction objectives (Teoh et al., 2025; Shi et al., 2026), while ConceptLM predicts discrete concepts that guide subsequent token generation (Liu et al., 2026). In contrast, other methods generate latent representations directly. Early work autoregressively generates compressed discrete latent sequences (Kaiser et al., 2018); LCM and CALM autoregress over sentence- or chunk-level continuous representations (LCM team et al., 2024; Shao et al., 2025); and latent diffusion or flow methods generate continuous representations before decoding them into text (Lovelace et al., 2023; Jiang et al., 2026; Hu et al., 2026; Guo et al., 2026; Liang et al., 2026). These works either use latent representations to support token generation or generate latents along sequence positions or diffusion trajectories. AXIS^2 instead makes representation resolution the autoregressive axis, generating progressively more informative representations of the same target sequence.

Discrete Latent Hierarchies and Scale-Wise Generation. Hierarchical computation has been explored through multiscale patches and dynamic chunking (Yu et al., 2023; Pagnoni et al., 2025; Hwang et al., 2026), while discrete autoencoders such as VQ-VAE and residual quantization provide learned latent code spaces (van den Oord et al., 2017; Lee et al., 2022). MaskGIT introduces iterative refinement in discrete representations (Chang et al., 2022), and VAR reformulates image autoregression as coarse-to-fine next-scale prediction (Tian et al., 2024), with later work incorporating text conditioning and intra-scale refinement (Ma et al., 2024; Kumbong et al., 2025). While these works demonstrate the value of discrete hierarchies and scale-wise generation, they primarily study hierarchical representations, multiscale architectures, or coarse-to-fine generation in vision. AXIS^2 instead uses representation resolution as a generative axis for language and introduces progressive within-resolution commitment as a second axis for dependency modeling.

3 AXIS^2: TWO-DIMENSIONAL LATENT GENERATION

Most language generation methods organize generation along a single progression. Autoregressive models advance over token positions, while masked-token and discrete diffusion models typically refine variables iteratively at a fixed token resolution. AXIS^2 instead organizes generation over a two-dimensional state space in latent representation space.

3.1 HIERARCHICAL MULTI-RESOLUTION REPRESENTATIONS

Let $\mathbf{x}$ denote a target sequence. A pretrained and frozen tokenizer maps $\mathbf{x}$ into a hierarchy of discrete latent codes $\mathcal{Q} = (\mathbf{q}_1, \ldots, \mathbf{q}_K)$, where level k contains L_k latent positions, with $L_1 < \cdots < L_K$, and each position is represented by S product-quantization segments.

The levels in $\mathcal{Q}$ should not be interpreted as independent semantic representations. Instead, they provide *incremental information* at progressively finer representation resolutions. We instantiate this hierarchy using residual quantization. Let $\mathbf{f}_k$ denote the accumulated latent representation after incorporating levels $1{:}k$. Then $\mathbf{f}_k = \mathbf{f}_{k-1} + U_k(D_k(\mathbf{q}_k))$, where D_k dequantizes the discrete codes at level k, and U_k maps them to the common latent resolution. Thus, $\mathbf{q}_k$ specifies the residual information introduced at resolution k, whereas $\mathbf{f}_k$ represents all information accumulated up to that resolution.

The sequence $\mathbf{f}_1 \to \mathbf{f}_2 \to \cdots \to \mathbf{f}_K$, therefore forms a progressively more informative representation of the target. Early accumulated representations retain only part of the information required to reconstruct $\mathbf{x}$, while later levels add residual information that resolves increasingly fine details. The discrete variables $(\mathbf{q}_1, \ldots, \mathbf{q}_K)$ are the quantities generated by AXIS^2 to traverse this hierarchy.

This representation hierarchy provides the first dimension of the AXIS^2 generation space. Rather than committing directly to the final token sequence, the model can progressively move from a partial representation toward one containing sufficient information for final text reconstruction.

3.2 AXIS I: REPRESENTATION RESOLUTION

The first axis advances generation across representation resolutions. Starting from an initial representation $\mathbf{f}_0$, AXIS^2 progressively constructs increasingly informative representations by generating the residual codes at each level, $\mathbf{f}_0 \xrightarrow{\mathbf{q}_1} \mathbf{f}_1 \xrightarrow{\mathbf{q}_2} \cdots \xrightarrow{\mathbf{q}_K} \mathbf{f}_K$. The corresponding latent variables are factorized as

$$p_\theta(\mathcal{Q} \mid \mathbf{c}) = \prod_{k=1}^{K} p_\theta(\mathbf{q}_k \mid \mathbf{q}_{<k}, \mathbf{c}), \tag{1}$$

where $\mathbf{q}_{<k}$ determines the accumulated representation $\mathbf{f}_{k-1}$ defined in Section 3.1.

At resolution k, the model does not regenerate the representation from scratch. Instead, it predicts the residual codes $\mathbf{q}_k$ conditioned on the information already established at coarser resolutions. These codes introduce an additional information increment, $\Delta\mathbf{f}_k = U_k(D_k(\mathbf{q}_k))$,, which updates the accumulated representation as $\mathbf{f}_k = \mathbf{f}_{k-1} + \Delta\mathbf{f}_k$. Progression along Axis I therefore preserves information established at earlier resolutions while successively adding information that was not yet represented. Generation evolves from an under-specified representation toward one containing sufficient detail for final text reconstruction. Later stages operate on an already structured representation and only need to supply the remaining information rather than construct the output from scratch.

This changes the primary progression of generation from traversing output positions to traversing representation resolutions. Crucially, the expensive backbone dependency chain is indexed by representation resolution rather than directly by output length. Autoregressive decoding requires a sequential full-model dependency chain that grows with the number of output tokens T. AXIS^2, in contrast, advances the backbone along a substantially shorter resolution hierarchy, while many latent variables associated with the same resolution can be processed in parallel. This shifts expensive sequential computation from the token level to the representation-resolution level.

Fully parallel prediction within a representation level, however, would prevent variables in the same $\mathbf{q}_k$ from conditioning on one another. Axis II (Section 3.3) introduces a second generation dimension

to recover these within-resolution dependencies without reverting to fully token-level sequential decoding.

3.3 AXIS II: WITHIN-RESOLUTION COMMITMENT

Axis I allows many variables at the same representation resolution to be predicted in parallel, but fully parallel prediction prevents variables within that resolution from conditioning on one another. At the other extreme, resolving them one by one would recover a fully sequential generation process and sacrifice much of the efficiency gained from the resolution hierarchy. AXIS^2 therefore introduces a second generation axis that progressively commits variables within each representation level.

Let $\mathcal{I}_k$ denote the set of variables in $\mathbf{q}_k$. During generation, AXIS^2 maintains a sequence of increasingly committed subsets, $\varnothing = \mathcal{M}_{k,0} \subset \mathcal{M}_{k,1} \subset \cdots \subset \mathcal{M}_{k,R_k} = \mathcal{I}_k$, where R_k is the number of within-resolution decoding rounds. The variables newly committed at round r are $\mathcal{A}_{k,r} = \mathcal{M}_{k,r} \setminus \mathcal{M}_{k,r-1}$. Conditioned on the representations established at previous resolutions, generation within resolution k proceeds through a sequence of conditional predictions,

$$\mathbf{q}_{k,\mathcal{A}_{k,r}} \sim p_\phi\big(\cdot \mid \mathbf{h}_k, \mathbf{q}_{k,\mathcal{M}_{k,r-1}}\big)\,, \tag{2}$$

where $\mathbf{h}_k$ denotes the resolution-level context produced by the backbone. The commitment rule determines which predicted variables are added to $\mathcal{M}_{k,r}$ at each round.

This second axis progressively exposes within-resolution dependencies: variables predicted in later rounds can condition on those committed earlier at the same representation resolution. When $R_k = 1$, the entire level is predicted in parallel. Increasing R_k allows more predictions to condition on previously established variables, providing a controllable trade-off between parallelism and dependency modeling without reverting to fully token-level autoregression.

Crucially, within-resolution refinement is decoupled from the expensive resolution-level backbone computation. At resolution k, the backbone first computes $\mathbf{h}_k = F_\theta(\mathbf{q}_{<k}, \mathbf{c})$, which depends only on information established before the current resolution. Because current-resolution codes are not fed back into the backbone, $\mathbf{h}_k$ remains fixed throughout within-resolution generation and can be reused across all commitment rounds.

A lightweight refinement module then updates its predictions conditioned on the cached backbone representation and the current commitment state, $p_\phi\big(\mathbf{q}_{k,\mathcal{A}_{k,r}} \mid \mathbf{h}_k, \mathbf{q}_{k,\mathcal{M}_{k,r-1}}\big)$. Thus, additional commitment rounds introduce only lightweight refinement computation rather than additional full-backbone evaluations. Axis I advances the expensive backbone across representation resolutions, whereas Axis II resolves dependencies within a fixed resolution using the lightweight sampler.

3.4 LEARNING THE TWO-DIMENSIONAL GENERATOR

Having defined the inference process, we train AXIS^2 on states drawn from the same two-dimensional generation space. A training state is specified by $(k, \mathcal{M}_k)$, where k identifies the current representation resolution along Axis I, and $\mathcal{M}_k \subseteq \mathcal{I}_k$ specifies the variables already committed within that resolution along Axis II.

Given a target sequence $\mathbf{x}$, the frozen tokenizer deterministically produces its representation hierarchy $\mathcal{Q} = (\mathbf{q}_1, \ldots, \mathbf{q}_K)$. For each representation resolution k, the backbone conditions on the representations established at earlier resolutions and produces $\mathbf{h}_k = F_\theta(\mathbf{q}_{<k}, \mathbf{c})$. This trains the resolution-level generator to advance along Axis I: information accumulated at resolutions $< k$ is used to predict the residual information introduced by $\mathbf{q}_k$.

To train Axis II, we sample a valid commitment state $\mathcal{M}_k \subseteq \mathcal{I}_k$ and define the unresolved variables as $\mathcal{U}_k = \mathcal{I}_k \setminus \mathcal{M}_k$. The lightweight refinement module predicts these unresolved variables conditioned on both the cached backbone representation and the committed variables. For one commitment state, we optimize the normalized prediction loss

$$\mathcal{L}_k = -\frac{1}{|\mathcal{U}_k|} \sum_{i \in \mathcal{U}_k} \log p_\phi(q_{k,i} \mid \mathbf{h}_k, \mathbf{q}_{k,\mathcal{M}_k})\,. \tag{3}$$

These partially committed states mirror the intermediate states encountered during decoding, training the same refinement module to operate at different points along the within-resolution axis.

To improve supervision efficiency, we reuse each expensive backbone representation across multiple valid commitment states. Specifically, for each backbone forward, we sample $\left\{\mathcal{M}_k^{(m)}\right\}_{m=1}^{M}$, and evaluate the lightweight refinement module on all M states while sharing the same cached $\mathbf{h}_k$. The corresponding resolution-level objective is $\bar{\mathcal{L}}_k = \frac{1}{M}\sum_{m=1}^{M}\mathcal{L}_k^{(m)}$. Here, M controls the number of training commitment states reused per backbone representation; it is independent of the number of refinement rounds R_k used during inference. This increases the amount of supervision obtained from each expensive backbone computation without lengthening the inference trajectory.

Finally, the overall objective aggregates losses across representation resolutions. A flattened objective implicitly gives greater weight to resolutions containing more latent variables, which need not reflect their prediction difficulty. We therefore use a resolution-aware weighting, $\mathcal{L}_{\text{res}} = \sum_{k=1}^{K}\omega_k(\alpha)\,\bar{\mathcal{L}}_k$, where $\omega_k(\alpha)$ controls the relative contribution of different representation resolutions. We provide the specific weighting construction, commitment-state sampling procedure, and other optimization details in Appendix **??**.

3.5 Experimental Setup

Datasets. We train all models on OpenWebText2 (OWT2), the OpenWebText2 component of The Pile (Gao et al., 2020). We evaluate text continuation on four benchmarks: English Wikipedia[1](https://huggingface.co/datasets/wikimedia/wikipedia), English WikiSource[2](https://huggingface.co/datasets/wikimedia/wikisource), TinyStories (Eldan & Li, 2023), and One Billion Words (1BW) (Chelba et al., 2014). We evaluate all methods on a fixed set of 1,000 examples per benchmark, with separate held-out examples used for decoding configuration selection. Additional preprocessing, context-length, and generation details are provided in the Appendix **??**.

Baselines. We compare against representative language generation paradigms: (1) **Autoregressive LMs**: a GPT-2-style autoregressive Transformer (AR) (Radford et al., 2019); (2) **Masked-generation LMs**: CMLM with Mask-Predict (Ghazvininejad et al., 2019); (3) **Discrete diffusion LMs**: MDLM (Sahoo et al., 2024) and BD3-LM (Arriola et al., 2025); and (4) **Continuous and latent-space generative LMs**: SSD-LM (Han et al., 2023) and TextLDM (Jiang et al., 2026). For iterative 2B baselines, we retain the full-NFE MDLM and BD3-LM results and add their pre-specified low-NFE operating points. Low-cost settings are selected by inference budget rather than test-set quality. Detailed baseline configurations are provided in the Appendix **??**.

Metrics. We evaluate models along four dimensions. For overall generation quality, we report MAUVE (Pillutla et al., 2021) and generation perplexity (Gen-PPL). To assess diversity and degeneration, we use unigram entropy, Distinct-1/2 (Li et al., 2016), and 4-gram repetition. We measure similarity to the reference continuation using ROUGE-1/2/L (Lin, 2004) and BERTScore (Zhang et al., 2020). Finally, we evaluate generation efficiency using the number of model evaluations (NFE), latency, and throughput. Detailed definitions and evaluation settings are provided in the Appendix **??**.

Model Configurations. For controlled comparisons, all generative backbones use a 12-layer Transformer with hidden dimension 768 and 12 attention heads, corresponding to approximately 85M non-embedding parameters. Our main AXIS^2 configuration uses two-dimensional position–segment refinement with $M = 8$, C2K2 sampling, CFG, and scale reweighting with $\alpha = 0.25$. We additionally report its standard token-weighted counterpart to expose the overlap–fluency trade-off. All models are trained under matched generative token budgets where applicable. Further architecture, training, and decoding details are provided in the Appendix.

3.6 Main results: quality

Table **??** reports two registered AXIS^2 settings under the controlled 2B-token budget rather than combining their metric-wise maxima. The standard segment-refinement setting is the MAUVE-oriented point, whereas adding $\alpha = 0.25$ increases ROUGE-1 and ROUGE-2 on all four benchmarks

[1] [https://huggingface.co/datasets/wikimedia/wikipedia

[2] [https://huggingface.co/datasets/wikimedia/wikisource

Table 1: Generation quality under a 51.2B-token training budget. R1, R2, and RL denote ROUGE-1/2/L F1 ($\times 100$); BS denotes BERTScore-F1 ($\times 100$); and M denotes MAUVE@256. All methods are evaluated on the same 1,000-example continuation benchmarks. The best and second-best results are **bold** and underlined, respectively. MAUVE is not ranked on TinyStories and 1BW due to its floor effect on short reference sequences.

Model	Wikipedia					WikiSource				
	R1	R2	RL	BS	M	R1	R2	RL	BS	M
AR	18.63	1.71	10.11	75.42	0.049	22.13	1.92	11.56	76.50	0.031
CMLM	10.86	1.77	9.37	72.58	0.008	12.64	2.62	10.59	73.65	0.009
BD3-LM	24.09	2.63	12.29	79.21	**0.226**	28.64	3.05	13.99	**80.25**	**0.240**
AXIS^2	**29.06**	**4.82**	**15.00**	**79.36**	0.195	**31.26**	**4.90**	**15.55**	79.91	0.091

Model	TinyStories					1BW				
	R1	R2	RL	BS	M	R1	R2	RL	BS	M
AR	15.89	0.88	10.02	77.05	0.006	7.21	0.21	6.29	79.13	0.005
CMLM	15.62	2.33	11.88	78.06	0.005	9.38	**0.75**	8.95	78.87	0.005
BD3-LM	27.10	2.79	15.18	83.14	0.008	**10.97**	0.62	**9.35**	**82.52**	0.005
AXIS^2	**33.68**	**5.74**	**18.73**	**83.28**	0.007	10.23	0.51	8.75	81.88	0.005

and reduces Gen-PPL on three of them. The latter outperforms AR and MDLM in ROUGE-1 and ROUGE-2 across all four benchmarks. Compared with full-NFE BD3-LM, it improves ROUGE-1/2 from 23.21/2.29 to 24.99/2.92 on Wikipedia, from 27.53/2.67 to 30.39/3.41 on WikiSource, and from 25.27/2.09 to 29.25/2.71 on TinyStories, while BD3-LM remains slightly stronger on 1BW. CMLM, SSD-LM, and Ldiff, as well as the 2B TextLDM configuration, exhibit degenerate generation patterns under our evaluation protocol; we therefore retain them for completeness but do not interpret their ROUGE or Gen-PPL scores independently of distributional and diversity metrics. We also report the pre-specified low-cost MDLM-22 and BD3-LM-64 points to expose inference-budget sensitivity, but retain the full-NFE rows so that low-step degradation is not mistaken for each baseline's best attainable quality. BD3-LM-64 is outside that model's intended sampling regime and is treated only as a diagnostic.

At 12.8B tokens, we report both the registered token-weighted configuration and the selected full method rather than combining their metric-wise maxima. The token-weighted configuration gives AXIS^2's strongest ROUGE-1 on Wikipedia, WikiSource, and TinyStories. Adding scale reweighting improves ROUGE-2 on all four benchmarks and reduces Gen-PPL from 338.1/271.2/139.5/975.3 to 253.9/236.3/120.1/685.3, while slightly reducing ROUGE-1 on the first three datasets. Relative to the strict 2B result, both configurations change the within-scale refinement scheme, so the comparison is not a pure data-scaling result.

The selected full configuration outperforms AR and MDLM in ROUGE-1 and ROUGE-2 across all four benchmarks. It also exceeds TextLDM on every such comparison except TinyStories ROUGE-1, and exceeds BD3-LM on Wikipedia, WikiSource, and TinyStories while using 22 rather than 1,024 backbone evaluations. BD3-LM retains higher MAUVE on these datasets and higher ROUGE-1/2 on 1BW, exposing a complementary quality–efficiency trade-off. Thus, AXIS^2 occupies a distinct operating point from autoregressive, discrete-diffusion, and continuous-space generation methods.

3.7 Main results: efficiency

3.8 Content-Adaptive Representation Across Resolutions

As shown in Figure **??**, we analyze 2,048 OpenWebText2 validation sequences of 1,024 tokens using the 100M AXIS^2 model, with 300M used for robustness checks. The frozen tokenizer represents each sequence at 11 resolutions, $\{1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024\}$, where $\mathbf{q}_\ell$ contains ℓ latent positions, from the coarsest $\mathbf{q}_1$ to the finest $\mathbf{q}_{1024}$. We measure resolution difficulty by the cross-entropy of predicting $\mathbf{q}_\ell$ conditioned on ground-truth coarser representations $\mathbf{q}_{<\ell}$, and independently measure text difficulty using the per-token NLL of a frozen GPT-2-large model.

Obs. ❶: Prediction difficulty differs across resolutions. The left panel of Figure **??** shows a clear resolution-dependent difficulty profile. For the 100M model, prediction cross-entropy increases from 1.50 bits/slot at q_1 to 5.20 at q_{16} and 5.28 at q_{32}, before decreasing to 1.90 at q_{1024}. The 300M model exhibits the same overall pattern, with the highest difficulty again concentrated at intermediate resolutions. This shows that different resolutions are not interchangeable instances of the same prediction problem.

Obs. ❷: Resolution-wise prediction difficulty is content-adaptive. The middle panel of Figure **??** correlates the prediction loss at each resolution with the independently measured text difficulty using Pearson's correlation coefficient r. Intermediate resolutions q_8–q_{32} become harder as the underlying text becomes more difficult, with the strongest positive correlation at q_{16} ($r = 0.31$), whereas the finest resolution q_{1024} shows the opposite trend ($r = -0.32$). The same qualitative pattern persists for the 300M model, where q_{1024} remains negatively correlated with text difficulty ($r = -0.33$). This indicates that the prediction burden across resolutions is not fixed, but shifts systematically with the difficulty of the text.

Obs. ❸: Different resolutions contribute differently to the final text. The right panel of Figure **??** provides a complementary reconstruction view. We select six sequences spanning the external difficulty range and progressively decode increasingly fine prefixes of their ground-truth latent hierarchies. Using representations up to q_{32} recovers only about 10% of tokens exactly. Accuracy increases to 36% at q_{128} and 95% at q_{512}, while q_{1024} provides the remaining corrections and brings reconstruction close to 100%. The largest reconstruction gain occurs between q_{128} and q_{512}, showing that information is accumulated non-uniformly across resolutions.

Overall, these observations reveal a form of *content-adaptive representation* in AXIS^2: the resolution hierarchy is fixed, but the prediction burden carried by different resolutions changes with the difficulty of the underlying text, while different resolutions make distinct contributions to the final output.

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A APPENDIX

You may include other additional sections here.